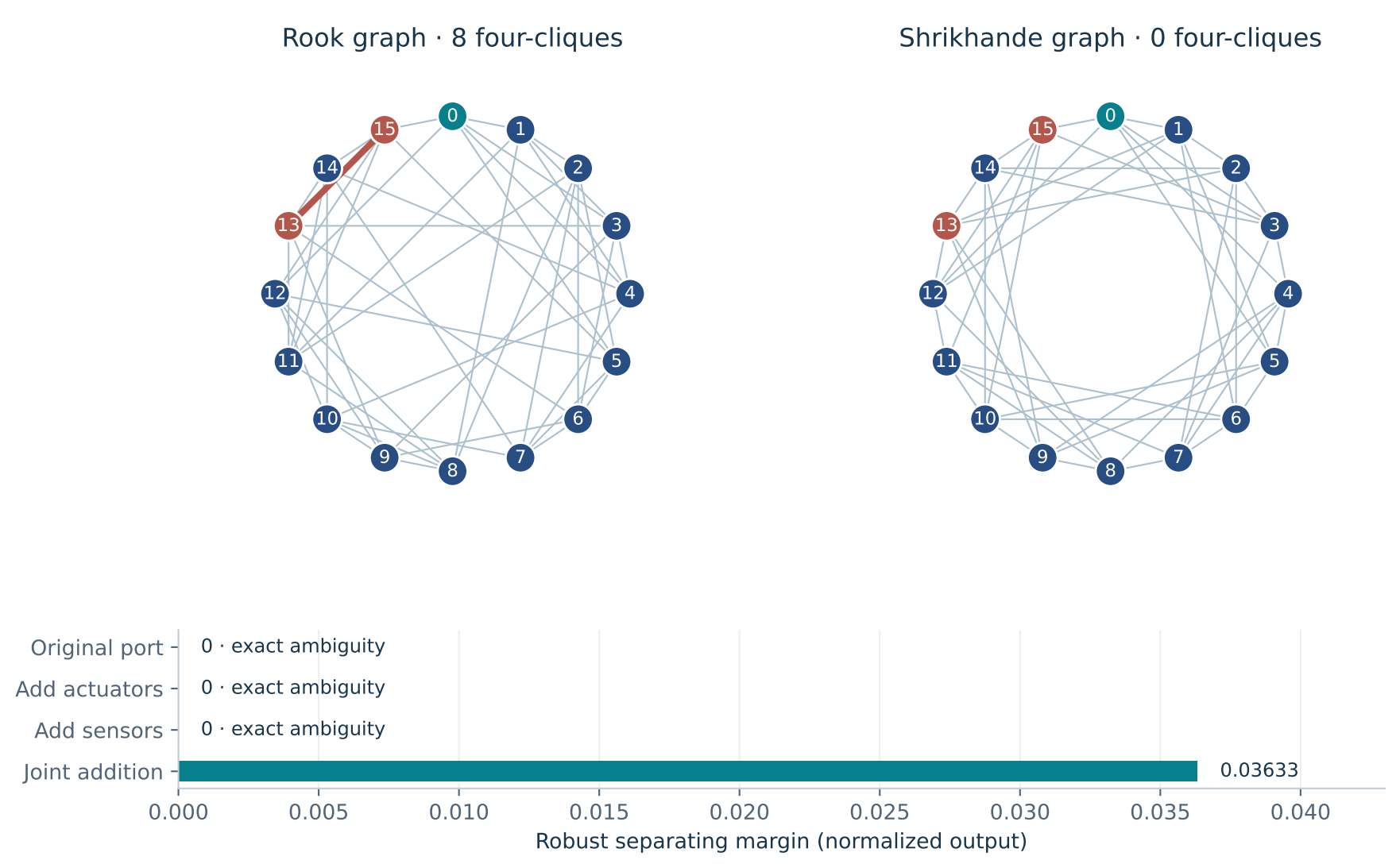The missing observable requires both new capabilities